

PROBLEM SHEET 1

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1. Write down a derivation of the judgement

$\text{succ}(\text{succ}(\text{succ}(\text{zero}))) \text{ odd}$

Solution:

$$\begin{array}{c}
 \frac{}{\text{zero even}} \text{ EVENZ} \\
 \frac{}{\text{succ}(\text{zero}) \text{ odd}} \text{ ODD} \\
 \frac{}{\text{succ}(\text{succ}(\text{zero})) \text{ even}} \text{ EVEN} \\
 \frac{}{\text{succ}(\text{succ}(\text{succ}(\text{zero}))) \text{ odd}} \text{ ODD}
 \end{array}$$

2. (i) Write down the rules that generate lists of natural numbers.
 (ii) Write down the associated induction principle.
 (iii) In your notation, write a derivation of the judgement that $[\mathbf{0}, \mathbf{1}]$ is a list.

Solution:

(i)

$$\frac{\text{NIL}}{\text{nil list}}$$

$$\frac{\text{CONS} \quad \frac{n \text{ nat} \quad xs \text{ list}}{\text{cons}(n, xs) \text{ list}}}{}$$

This is of course a mathematical restatement of the Haskell data types

```
data Nat = Zero | Succ Nat
data NatList = Nil | Cons Nat NatList
```

(ii) Let $\mathcal{P}$ be a property of the lists. If

- $\mathcal{P}(\text{nil})$, and
- whenever $n \text{ nat}$ and $\mathcal{P}(xs)$ we know that $\mathcal{P}(\text{cons}(n, xs))$

then $\mathcal{P}(xs)$ for all $xs \text{ list}$.

(There is also another principle that additionally inducts on numbers.)

(iii)

$$\begin{array}{c}
 \frac{}{\text{zero nat}} \text{ZERO} \qquad \frac{}{\text{succ(zero) nat}} \text{Succ} \qquad \frac{}{\text{nil list}} \text{NIL} \\
 \frac{}{\text{zero nat}} \text{ZERO} \qquad \frac{\text{succ(zero) nat} \quad \text{nil list}}{\text{cons(succ(zero), nil) list}} \text{CONS} \\
 \hline
 \text{cons(zero, cons(succ(zero), nil)) list} \quad \text{CONS}
 \end{array}$$

3. (*) All the judgements we have seen up to this point have been *unary*, in the sense that they referred to only one entity. For example, the judgement $n \text{ nat}$ only refers to the object n .

However, judgements can have arbitrary *arity*, and can thus define arbitrary relations between an arbitrary number of objects. For example, the following *ternary* judgment $\text{sum}(a, b, c)$ defines a relation between three objects: a , b and c .

$$\begin{array}{c}
 \text{BASE} \\
 \frac{b \text{ nat}}{\text{sum(zero, } b, b)}
 \end{array}$$

$$\begin{array}{c}
 \text{IND} \\
 \frac{\text{sum}(a, b, c)}{\text{sum(succ}(a), b, \text{succ}(c))}
 \end{array}$$

The judgement $\text{sum}(a, b, c)$ can be written in more familiar notation as $a + b = c$.

Such judgements can be used—amongst countless other things—to define functions. This exercise is about showing that the above rules define the addition function.

- (i) Write down a derivation of $\text{sum}(\text{succ}(\text{zero}), \text{succ}(\text{zero}), \text{succ}(\text{succ}(\text{zero})))$.
- (ii) Restate the above rules as a Haskell function on the data type

```
data Nat = Zero | Succ Nat
```

Does your code use pattern matching? Discuss its relation to the rules given above.

- (iii) Prove that if $\text{sum}(a, b, c)$ then $a \text{ nat}$, $b \text{ nat}$, and $c \text{ nat}$.
- (iv) (Existence) Prove that if $a \text{ nat}$ and $b \text{ nat}$ then there exists a $c \text{ nat}$ such that $\text{sum}(a, b, c)$.
- (v) (Uniqueness) Prove that if $\text{sum}(a, b, c)$ and $\text{sum}(a, b, c')$ it must be that $c = c'$.
- (vi) Conclude that $\text{sum}(a, b, c)$ indeed defines a function on natural numbers.

Solution:

```
(ii)      sum :: Nat -> Nat -> Nat
          sum Zero b = b
          sum (Succ a) b = Succ (sum a b)
```

(iii) We prove the claim by induction on the derivation of $\text{sum}(a, b, c)$.

Case(**BASE**). If the derivation is of the form

$$\frac{\frac{\vdots}{b \text{ nat}}}{\text{sum}(\text{zero}, b, b)} \text{BASE}$$

then we know (i) that zero nat by the rule **ZERO**, and (ii) that $b \text{ nat}$, because it is a premise of the rule **BASE**. So all three of zero , b , and b are natural numbers.

Case(**IND**). If the derivation is of the form

$$\frac{\frac{\vdots}{\text{sum}(a, b, c)}}{\text{sum}(\text{succ}(a), b, \text{succ}(c))} \text{IND}$$

then **by the inductive hypothesis (IH)** we know that $a \text{ nat}$, $b \text{ nat}$, and $c \text{ nat}$. Then,

- As $a \text{ nat}$, by the rule **SUCC** of natural numbers we deduce that $\text{succ}(a) \text{ nat}$.
- As $c \text{ nat}$, by the rule **SUCC** of natural numbers we deduce that $\text{succ}(c) \text{ nat}$.

Therefore, all three of $\text{succ}(a)$, b and $\text{succ}(c)$ are natural numbers.

(iv) We prove the claim by induction on the derivation of $a \text{ nat}$.

Case(**ZERO**). If the derivation is of the form

$$\overline{\text{zero nat}}$$

(so in fact $a \stackrel{\text{def}}{=} \text{zero}$) then we can use the rule **BASE** to prove that

$$\overline{\text{sum}(\text{zero}, b, b)}$$

Hence, there *does* exist a c such that $\text{sum}(a, b, c)$, and that c is in fact b .

Case(**SUCC**). If the derivation is of the form

$$\frac{\frac{\vdots}{x \text{ nat}}}{\text{succ}(x) \text{ nat}} \text{SUCC}$$

(which is to say that $a = \text{succ}(x)$ for some x). Then, by the **IH** applied to $x \text{ nat}$, we know that there exists a c' such that $\text{sum}(x, b, c')$. Given a

derivation of this judgement, we use the rule **IND** to deduce that

$$\frac{\frac{\vdots}{\text{sum}(x, b, c')}}{\text{sum}(\text{succ}(x), b, \text{succ}(c'))} \text{IND}$$

Recalling that $a = \text{succ}(x)$, we see that there *does* exist a c so that $\text{sum}(a, b, c)$, namely $c \stackrel{\text{def}}{=} \text{succ}(c')$.

Note: Notice that this proof is essentially just a regular natural number induction on a . Also, notice that it is very similar to the function definition given in Haskell above.

(v) We prove the claim by induction on the derivation of $\text{sum}(a, b, c)$.

Case(**BASE**). Suppose the derivation is of the form

$$\overline{\text{sum}(\text{zero}, b, b)}$$

(which is to say that $a = \text{zero}$ and $c = b$). By assumption we also know that $\text{sum}(\text{zero}, b, c')$ for some c' . As the first component of this judgement is a zero , its derivation can only be of the form

$$\overline{\text{sum}(\text{zero}, b, b)}$$

(no other rule has a zero in the first component!). Thus, $c' = b = c$.

Case(**IND**). Suppose the derivation is of the form

$$\frac{\frac{\vdots}{\text{sum}(x, b, y)}}{\text{sum}(\text{succ}(x), b, \text{succ}(y))} \text{IND}$$

(which is to say that $a = \text{succ}(x)$ and $c = \text{succ}(y)$ for some x and y). Consider the derivation of $\text{sum}(a, b, c')$. As $a = \text{succ}(x)$, this can only be of the form

$$\frac{\frac{\vdots}{\text{sum}(x, b, y')}}{\text{sum}(\text{succ}(x), b, \text{succ}(y'))} \text{IND}$$

for some y' (no other rule matches the shape!). So, in fact, $c' = \text{succ}(y')$.

At this point we have derivations of $\text{sum}(x, b, y)$ and $\text{sum}(x, b, y')$. By the **IH**, we obtain that $y = y'$. Hence $c = \text{succ}(y) = \text{succ}(y') = c'$.